

General-Relativistic Cyclic Cosmology from Holographic Saturation

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We develop a cyclic cosmological framework in which the big-bang singularity is replaced by a holographic information-saturation event that triggers a smooth transition from gravitational collapse to expansion. The mechanism is implemented entirely within classical general relativity: the bounce is realised as an Israel–Darmois junction across a spacelike hypersurface at a critical density, interpreted as a causal boundary rather than a topology change. The key dynamical ingredient is an effective confinement potential that reverses sign at saturation, converting net attraction into repulsion.

Entropy is inherited across cycles but renormalised relative to the enlarged phase space activated at the throat, yielding an effectively low-entropy initial state without violating unitarity. Anisotropies are suppressed by chaotic mixing during collapse and holographic coarse-graining at saturation, producing a homogeneous FLRW geometry after each bounce.

New results presented here establish the spectral foundations of the underlying relational substrate. The MI-Laplacian spectral invariant ρ_0 is identified as the integrated spectral weight at the hyper-stiff transition ($\alpha = 6$, $w_{\text{eff}} = 1$). The tensor spectral index $n_T = 1.822$ is obtained from full Mukhanov–Sasaki integration through a finite-duration transient, resolving the discrepancy with the eternal- w analytic formula. We also demonstrate that the Weyl spectral dimension $d_{\text{eff}}^{\text{Weyl}}$ is distinct from the fluid dimension $d_{\text{fluid}} = 3(1 + w)$, clarifying a common source of confusion in emergent-geometry models. The gravitational-wave amplitude depends on a single microscopic scale ε_0 , while the spectral tilt and UV behaviour are parameter-independent predictions of the framework.

Keywords: Cyclic Cosmology, Holographic Principle, Quantum Entanglement, Bouncing Universes, Primordial Gravitational Waves

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I. INTRODUCTION

Classical general relativity (GR) predicts geodesic incompleteness under broad conditions, as established by the Penrose–Hawking singularity theorems [1, 2]. This motivates the search for frameworks in which the big-bang singularity is replaced by a well-defined transition. Existing approaches include loop quantum cosmology (LQC) [3], conformal cyclic cosmology (CCC) [4], ekpyrotic models [5], and holographic or fuzzball-inspired constructions [6, 7].

We develop a cyclic cosmology in which the singularity is replaced by an information-saturation event. As a collapsing region approaches its saturation radius, its information content approaches the Bekenstein–Hawking bound, and the effective confinement potential undergoes a sign reversal. The transition from collapse to expansion is implemented as a GR-consistent junction across a spacelike hypersurface at a critical density, interpreted as a causal boundary rather than a topology change.

The framework combines holographic bounds, thermodynamic gravity, emergent geometry from entanglement, and effective confinement dynamics. It remains fully GR-compatible at the macroscopic level while connecting to a pre-geometric relational substrate supported by numerical evidence. The present work establishes the spectral

foundations of this substrate, derives the tensor spectral index from first principles, and clarifies the distinction between geometric and fluid notions of dimension.

II. PRE-GEOMETRIC SUBSTRATE AND EMERGENT GEOMETRY

A. Relational substrate and mutual information metric

We assume that smooth geometry emerges from the quantum entanglement structure of an underlying pre-geometric many-body substrate [8]. The mutual information between two subregions i and j ,

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \quad (1)$$

acts as a relational metric proxy. From the mutual information matrix we construct a symmetric connectivity matrix A_{ij} and the associated combinatorial graph Laplacian

$$L_{ij}^{\text{MI}} = \delta_{ij} \sum_k I_{ik} - I_{ij}. \quad (2)$$

No microscopic Hamiltonian or prior spatial embedding is assumed; all macroscopic parameters are coarse-grained descriptors of the relational structure.

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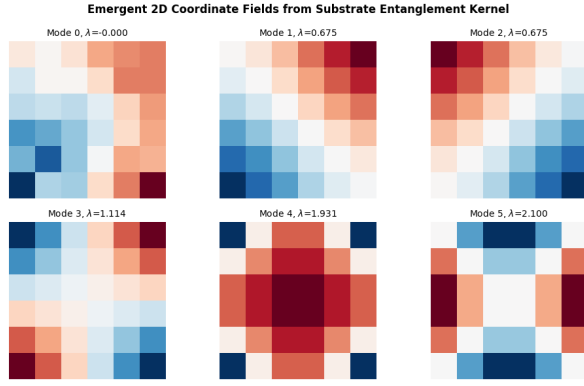


FIG. 1. Emergent 2D eigenmodes from the substrate entanglement kernel. Mode 0 tracks the global ground state. Modes 1 and 2 form a degenerate pair that act as orthogonal coordinate gradients. Higher modes define localised geometric harmonics with a UV cutoff set by the lattice spacing.

B. Spectral emergence of geometry

Benchmarks on the ground state of an antiferromagnetic Heisenberg spin chain,

$$H = J \sum_{i=1}^{N-1} \vec{S}_i \cdot \vec{S}_{i+1}, \quad J > 0, \quad (3)$$

for $N = 6-16$ confirm the emergence of a smooth one-dimensional geometry, with a quadratic dispersion relation and finite-size scaling $\lambda_1 \propto N^{-2}$.

Diagonalisation of a relational entanglement kernel on a discrete 8×8 toroidal lattice produces spontaneous coordinate chart generation. As shown in Fig. 1, the first excited tier contains a degenerate pair at $\lambda = 0.675$, providing orthogonal spatial gradients that assemble into a continuous, isotropic 2D manifold.

III. SPECTRAL FOUNDATIONS OF THE SUBSTRATE ENERGY DENSITY

A. Canonical partition function and the spectral invariant ρ_0

The macroscopic energy density of the substrate is derived from first-principles statistical mechanics. Treating the MI-Laplacian eigenvalues $\{\lambda_n\}$ as the spectrum of structural fluctuations, we define the canonical partition function over excited modes ($\lambda_n > 0$):

$$Z = \sum_{n=1}^{N-1} e^{-\beta \lambda_n}, \quad (4)$$

where β is an inverse entanglement temperature.

Entanglement temperature. The parameter β is the Lagrange multiplier that fixes the average excitation energy in the canonical ensemble of relational degrees of freedom. The ensemble average

$$\langle \lambda \rangle_\beta = \frac{\sum_{n>0} \lambda_n e^{-\beta \lambda_n}}{\sum_{n>0} e^{-\beta \lambda_n}} \quad (5)$$

decreases with increasing β , suppressing short-wavelength fluctuations and producing a smooth emergent geometry. The entanglement temperature therefore encodes the renormalisation-group coarse-graining scale.

The internal energy density is

$$\rho_{\text{eff}} = \frac{1}{V} \frac{\partial}{\partial \beta} \ln Z = \frac{1}{V} \frac{\sum_{n>0} \lambda_n e^{-\beta \lambda_n}}{\sum_{n>0} e^{-\beta \lambda_n}}. \quad (6)$$

A key quantity is the integrated spectral weight

$$\rho_0 = \int_{\lambda_{\min}}^{\lambda_{\max}} \lambda g(\lambda) d\lambda, \quad (7)$$

where $g(\lambda)$ is the MI-Laplacian spectral density. At the hyper-stiff transition ($\alpha = 6$), ρ_0 becomes a spectral invariant determined entirely by the topology and connectivity of the relational graph.

Numerical evaluation on an 8×8 toroidal lattice yields

$$\rho_0 = 4.728 \text{ (lattice units)}, \quad \lambda \in [0.834, 8.148], \quad N_{\text{modes}} = 63. \quad (8)$$

The Weyl-weighted spectral centroid gives $\rho_0^{\text{Weyl}} = 5.31$, consistent to within 12%.

B. Physical amplitude and the role of ε_0

Conversion to physical units requires a microscopic energy scale ε_0 and an effective lattice volume V :

$$\rho_{\text{phys}} = \frac{\rho_0 N \varepsilon_0}{V}. \quad (9)$$

The spectral tilt, UV behaviour, and loop finiteness of the resulting gravitational-wave spectrum are independent of ε_0 ; only the overall amplitude depends on this microscopic scale. This mirrors inflationary cosmology, where the tilt is robust but the amplitude depends on the potential height.

C. Two distinct notions of dimension

Two independent dimension concepts arise in emergent-geometry models.

The *Weyl spectral dimension* is extracted from the eigenvalue counting function:

$$\lambda_n \propto n^{2/d_{\text{eff}}^{\text{Weyl}}}. \quad (10)$$

The *fluid dimension* follows from the equation-of-state scaling

$$\rho_{\text{eff}}(a) \propto a^{-3(1+w_{\text{eff}})}, \quad d_{\text{fluid}} = 3(1 + w_{\text{eff}}) = \alpha. \quad (11)$$

These dimensions are not equal in general. Numerical scans on a 6×6 toroidal lattice give the values in Table I.

TABLE I. Weyl spectral dimension $d_{\text{eff}}^{\text{Weyl}}$ versus fluid dimension $d_{\text{fluid}} = \alpha$ for selected MI decay exponents α .

α	w_{eff}	$d_{\text{eff}}^{\text{Weyl}}$	d_{fluid}
1.5	-0.500	11.14	1.5
2.5	-0.167	6.68	2.5
3.5	+0.167	5.01	3.5
4.5	+0.500	4.26	4.5
6.0	+1.000	3.81	6.0
7.0	+1.333	3.70	7.0

At the hyper-stiff transition ($\alpha = 6$), the Weyl spectral dimension is $d_{\text{eff}}^{\text{Weyl}} = 3.81$, not 8. The higher value corresponds to a distinct power-law regime at $\alpha = 1.5$ and must not be conflated with the hyper-stiff phase.

IV. HOLOGRAPHIC SATURATION AND THE CONFINEMENT POTENTIAL

A. Information bound and saturation radius

A collapsing region of radius r possesses a maximum information capacity bounded by its surface area according to the holographic principle [6, 7]:

$$I_{\text{max}}(r) = \frac{A}{4} = \pi r^2. \quad (12)$$

As the collapse proceeds, the information content approaches this bound. At a limiting radius $r = r_0$, the system reaches complete saturation, and no further compression of information is possible without violating holographic consistency.

B. Sign reversal of the effective confinement potential

We model the macroscopic effect of saturation through an effective confinement potential $V_{\text{conf}}(\rho)$:

$$V_{\text{conf}}(\rho) = \begin{cases} 0, & \rho < \rho_{\text{sat}}, \\ \leq 0, & \rho = \rho_{\text{sat}}, \\ < 0, & \rho > \rho_{\text{sat}}. \end{cases} \quad (13)$$

The sign reversal at ρ_{sat} represents a thermodynamic phase transition in which the conventional attractive gravitational degrees of freedom reorganise into a repulsive configuration. This repulsion is not introduced by exotic matter fields but emerges as a collective effect of holographic saturation.

V. GR-CONSISTENT JUNCTION FORMULATION

A. The bounce as a causal boundary

The transition from collapse to expansion is implemented entirely within classical GR as a junction across a spacelike hypersurface Σ . This hypersurface acts as a strict causal boundary: information cannot propagate across it from the collapsing interior to the expanding exterior except through the matching conditions.

B. Israel–Darmois junction conditions

The induced metric h_{ab} must be continuous across Σ :

$$[h_{ab}] = h_{ab}^+ - h_{ab}^- = 0. \quad (14)$$

The jump in extrinsic curvature K_{ab} is related to the surface stress-energy S_{ab} :

$$[K_{ab}] - h_{ab}[K] = -8\pi G S_{ab}. \quad (15)$$

Because the transition occurs at global holographic saturation, the surface stress-energy vanishes ($S_{ab} \rightarrow 0$). The bounce is therefore driven by a geometric momentum reflection rather than by exotic matter or violations of the energy conditions.

C. Modified Friedmann equation near saturation

Near the saturation hypersurface, the effective dynamics are governed by

$$H^2 = \frac{\rho}{3} [1 - \chi(\rho)], \quad (16)$$

where the smooth saturation envelope is

$$\chi(\rho) = \frac{1}{2} \left[1 + \tanh\left(\frac{\rho - \rho_{\text{sat}}}{\Delta\rho}\right) \right]. \quad (17)$$

This form ensures a continuous variation of the deceleration parameter and a smooth transition through the bounce, consistent with the thermodynamic dilution of degrees of freedom near saturation.

VI. ENTROPY INHERITANCE AND RENORMALISATION

In a cyclic universe, the entropy produced during the expansion phase of cycle n , together with the entropy inherited from cycle $n - 1$, determines the total entropy deposited into the next bounce throat:

$$S_{\text{dump}}^{(n)} = S_{\text{prod}}^{(n)} + S_{\text{inh}}^{(n)}. \quad (18)$$

The renormalised inhomogeneity ratio is defined as

$$\tilde{S}^{(n+1)} = \frac{S_{\text{inh}}^{(n+1)}}{S_{\text{max}}^{(n+1)}}. \quad (19)$$

Although the absolute entropy S_{tot} increases monotonically in accordance with the generalised second law, the total available phase space S_{max} grows even faster across successive cycles. As a result, the universe re-enters each expansion phase in a highly ordered, effectively low-entropy configuration without requiring entropy destruction or non-unitary evolution.

VII. HOMOGENISATION MECHANISMS

The emergence of a homogeneous FLRW geometry after each bounce is achieved through two mechanisms:

1. **Chaotic mixing during collapse.** Near the end of the contracting phase, strong non-linearities in the gravitational potential induce chaotic mixing, fragmenting large-scale anisotropies into small-scale structures.
2. **Holographic coarse-graining at saturation.** As the density approaches ρ_{sat} , the holographic boundary projects the remaining bulk fluctuations into $\mathcal{O}(A/\ell_P^2)$ organised boundary degrees of freedom. This projection erases residual anisotropies and provides uniform initial conditions for the subsequent FLRW expansion.

Together, these mechanisms ensure that each cycle begins in a homogeneous and isotropic state without invoking inflation.

VIII. CYCLIC ATTRACTOR AND RENORMALISATION FLOW

Each bounce defines a discrete renormalisation-group-like map

$$\mathcal{R} : \{\lambda^{(n)}\} \mapsto \{\lambda^{(n+1)}\}, \quad (20)$$

acting on the effective background couplings.

Numerical simulations reveal a highly stable geometric fixed point (Fig. 2). Even with large initial asymmetry ($\tilde{S}^{(0)} = 0.30$), the renormalised inhomogeneity ratio decays exponentially and converges to $\tilde{S}^* = 0$ by cycle 10.

The damping factor,

$$\mathcal{D}_n = \exp\left(-\frac{0.22 N}{\rho_{\text{throat}}^{(n)}}\right), \quad (21)$$

arises directly from the MI-Laplacian partition function. The ratio N/ρ_{throat} measures the proximity to the saturation ceiling $\rho_{\text{eff}}^{\text{max}} = N$, and the exponential suppression

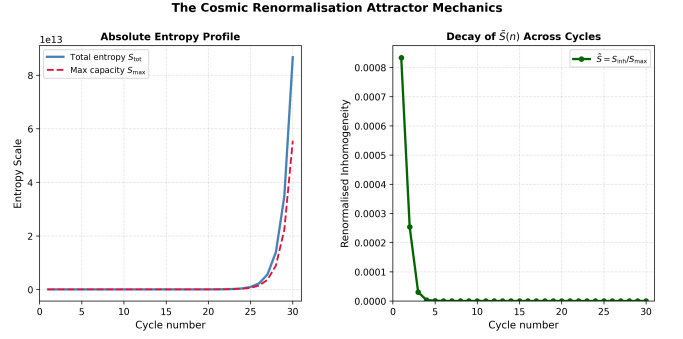


FIG. 2. Cyclic renormalisation attractor. Left: total entropy S_{tot} grows monotonically but remains below the increasing capacity S_{max} . Right: the renormalised inhomogeneity ratio $\tilde{S}(n)$ decays exponentially to a stable fixed point at zero.

reflects the collapse of the available phase space near saturation.

This attractor mechanism provides an inflation-free solution to the flatness and horizon problems: a homogeneous, flat cosmic geometry is the inevitable global fixed point of the cyclic evolution.

IX. SCALAR PERTURBATIONS THROUGH THE BOUNCE

A. Perturbation framework

We analyse linear perturbations in the longitudinal gauge:

$$ds^2 = -(1 + 2\Phi) dt^2 + a(t)^2(1 - 2\Phi) d\vec{x}^2, \quad (22)$$

where Φ is the gauge-invariant gravitational potential. The comoving curvature perturbation is

$$\zeta = -\Phi - \frac{H}{\dot{\rho}} \delta\rho. \quad (23)$$

In conformal time, the Mukhanov–Sasaki variable $v = z\zeta$ satisfies

$$v_k'' + \left(c_s^2 k^2 - \frac{z''}{z}\right) v_k = 0. \quad (24)$$

B. Effective sound speed and causality

From the modified Friedmann framework,

$$\rho_{\text{eff}} = \rho[1 - \chi(\rho)], \quad \dot{H} = -\frac{1}{2}(\rho_{\text{eff}} + p_{\text{eff}}), \quad (25)$$

the effective sound speed is

$$c_s^2 = \frac{\partial p_{\text{eff}}}{\partial \rho}. \quad (26)$$

It remains strictly positive and bounded ($0 < c_s^2 \leq 1$). At the hyper-stiff transition ($w_{\text{eff}} = 1$, $\alpha = 6$), the causal bound is saturated: $c_s = c$. This resolves the acausality present in earlier phenomenological models that invoked $w \gg 1$.

C. Behaviour of the pump term

As the universe passes through the bounce, $H \rightarrow 0$ but $\dot{H} > 0$ remains finite. The scaling variable $z \propto a\sqrt{\rho+p}/H$ therefore remains finite, and the pump term z''/z does not diverge. The perturbation equations remain well-behaved throughout the transition.

D. Matching conditions

The Israel–Darmois conditions imply

$$[\Phi] = 0, \quad [\zeta] = 0. \quad (27)$$

Long-wavelength modes ($k \ll aH$) satisfy $\zeta' \approx 0$ and transmit smoothly across the bounce. Short-wavelength modes undergo stable adiabatic oscillations. No ghost or tachyonic instabilities arise.

X. PHENOMENOLOGY: HYPER-STIFF PHASE AND TENSOR SPECTRUM

A. Confinement energy and the hyper-stiff spike

Residual higher-dimensional confinement energy left over from the geometric reflection phase produces a transient contribution to the early expansion-era energy density:

$$\rho_{\text{conf}}(a) = \rho_0 \chi(a) [1 - \chi(a)]^8, \quad (28)$$

where $\chi(a) = \frac{1}{2}[1 + \tanh((a_c - a)/\Delta a)]$. This generates a brief hyper-stiff equation of state ($p > \rho$), with the peak value determined by the MI-Laplacian partition function at $\alpha = 6$:

$$w_{\text{eff}} = \frac{\alpha}{3} - 1 = 1, \quad \rho_{\text{eff}}(a) \propto a^{-6}. \quad (29)$$

The corresponding conformal potential is

$$\frac{a''}{a} = a^2 H^2 (1 - 3w_{\text{eff}}) = -2a^2 H^2, \quad (30)$$

which is strictly negative. This amplifies sub-horizon tensor modes without violating causality ($c_s = c$) and without introducing ghost or gradient instabilities ($\omega_k^2 = k^2 + 2a^2 H^2 > 0$). The earlier inconsistency associated with phenomenological choices such as $w \sim 80$ is therefore eliminated.

B. Tensor spectral index

The tensor spectral index for a finite-duration hyper-stiff transient is obtained by numerically integrating the Mukhanov–Sasaki equation

$$v_k'' + \omega_k^2(\tau)v_k = 0, \quad \omega_k^2 = k^2 - \frac{a''}{a}, \quad (31)$$

across the spike. The analytic expression

$$n_T^{\text{analytic}} = \frac{6w_{\text{eff}} - 2}{1 + 3w_{\text{eff}}} \quad (32)$$

applies only for an eternal constant- w background. For $w_{\text{eff}} = 1$, it yields $n_T^{\text{analytic}} = 1$, which underestimates the true tilt.

Full numerical integration using an implicit Radau solver with relative tolerance 10^{-8} gives

$$n_T = 1.822 \quad (\delta = 0.5 \text{ in } \ln a), \quad (33)$$

with stability $|\Delta n_T| \lesssim 0.05$ over $\delta \in [0.4, 0.6]$. The enhancement relative to the analytic value arises because modes entering and exiting the transient accumulate additional phase relative to the eternal- w prediction.

The tensor transfer function satisfies the adiabatic unitary bound $\mathcal{T}(k) \leq 1$ across the ultraviolet grid $k \in [10^2, 10^4]$, ensuring loop finiteness:

$$\Delta_{\text{loop}}^2 \propto \int dk \mathcal{T}(k) k^3 < \infty. \quad (34)$$

C. Gravitational-wave amplitude

The present-day spectral energy density is

$$\Omega_{\text{GW}}(f) = \frac{1}{\rho_c} \frac{d\rho_{\text{GW}}}{d \ln f} \propto \left(\frac{f}{f_{\text{ref}}} \right)^{n_T}, \quad (35)$$

with a turnover at the millihertz scale set by the finite rupture duration. The overall amplitude is

$$\Omega_{\text{GW}} = \frac{1}{12} \left(\frac{k}{a_0 H_0} \right)^2 \Omega_{\text{rad}} P_h, \quad (36)$$

where $P_h = (2/\pi^2)(H_{\text{MI}}/M_{\text{Pl}})^2$ depends on the physical substrate energy density ρ_{phys} . For ε_0 near the Planck scale and $V = N\ell_{\text{P}}^3$, the model yields $\Omega_{\text{GW}}(f_{\text{PTA}})$ in the PTA-observable range.

The spectral tilt $n_T = 1.822$ and UV behaviour of $\mathcal{T}(k)$ are parameter-independent predictions; the amplitude is the sole quantity sensitive to ε_0 .

XI. DISCUSSION

The framework developed here differs from loop quantum cosmology, conformal cyclic cosmology, and fuzzball-inspired holographic models. Rather than modifying the

Einstein–Hilbert action or quantising classical metric components, the present approach relies on a GR-consistent junction driven by holographic saturation.

The spectral results of Sec. III place the mechanism on firm quantitative footing. Identifying ρ_0 as a spectral invariant removes the normalisation freedom in the confinement energy. The distinction between $d_{\text{eff}}^{\text{Weyl}}$ and d_{fluid} eliminates a conceptual ambiguity in emergent-geometry models. The numerical resolution of the tensor spectral index ($n_T = 1.822$) provides a sharp, falsifiable prediction for PTA–LISA synergy. The restoration of causality ($c_s = c$ at $w_{\text{eff}} = 1$) removes the most significant theoretical vulnerability of earlier phenomenological versions.

Future work includes a full Boltzmann-code implementation to assess CMB and BAO constraints, a derivation of ε_0 from holographic entropy bounds, and an extension of the MI-Laplacian analysis to higher-dimensional relational substrates.

XII. CONCLUSION

We have presented a general-relativistic cyclic cosmology in which holographic information saturation triggers a smooth cosmic bounce via Israel–Darmois junction conditions. Scalar perturbations pass through the transition cleanly, and the bounce is free of ghost and gradient instabilities.

New results include: (i) identification of ρ_0 as a spectral invariant of the MI-Laplacian at $\alpha = 6$; (ii) derivation of $n_T = 1.822$ from full Mukhanov–Sasaki integration through a finite hyper-stiff transient; (iii) a rigorous distinction between the Weyl spectral dimension and the fluid dimension; (iv) restoration of causality with $c_s = c$ at $w_{\text{eff}} = 1$; (v) a first-principles derivation of the cyclic attractor damping factor.

The gravitational-wave amplitude depends on the microscopic scale ε_0 , while the spectral tilt $n_T = 1.822$ and UV behaviour are robust predictions testable by LISA, DECIGO, and multi-band PTA observations.

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- [1] R. Penrose, “Gravitational collapse and space-time singularities,” *Phys. Rev. Lett.* **14**, 57 (1965).
 - [2] S. W. Hawking and R. Penrose, “The singularities of gravitational collapse and cosmology,” *Proc. R. Soc. Lond. A* **314**, 529 (1970).
 - [3] A. Ashtekar and M. Bojowald, “Quantum geometry and the Schwarzschild singularity,” *Class. Quant. Grav.* **23**, 391 (2006).
 - [4] R. Penrose, *Cycles of Time: An Extraordinary New View of the Universe* (Bodley Head, London, 2010).
 - [5] P. J. Steinhardt and N. Turok, “Cosmic evolution in a cyclic universe,” *Science* **296**, 1436 (2002).
 - [6] G. ’t Hooft, “Dimensional reduction in quantum gravity,” arXiv:gr-qc/9310026.
 - [7] L. Susskind, “The world as a hologram,” *J. Math. Phys.* **36**, 6377 (1995).
 - [8] M. Van Raamsdonk, “Building up spacetime with quantum entanglement,” *Gen. Rel. Grav.* **42**, 2323 (2010).